

The Cost of Stability: A Comparative Evaluation of 6T, 8T and 10T SRAM Cells Across Technology Nodes

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Abstract—CMOS scaling and the rising on-die cache fraction of modern processors have made static random-access memory (SRAM) cell stability a first-order design constraint. The canonical six-transistor (6T) cell suffers from a well-known conflict between read stability (favored by a large pull-down/access ratio β) and write-ability (favored by a small pull-up/access ratio γ). This work *quantifies the cost of stability* across the 6T, 8T, and 10T topologies through five complementary figures-of-merit (FOMs): hold, read, and write static noise margins (HSNM/RSNM/WSNM); the static N -curve current and voltage margins (SINM, SVN); and the read access time T_{rd} . Each FOM is swept over (β, γ) and across seven process nodes from 180 nm down to 22 nm using the SkyWater 130 nm Open PDK and the Predictive Technology Model. Three results emerge. (i) HSNM and WSNM are properties of the shared storage core and write path: they are numerically identical across all three topologies at every node (HSNM falls from 460 mV at 180 nm/1.8 V to 235 mV at 22 nm/0.8 V for all cells). (ii) The 6T cell loses 27–67% of its hold margin under read at $\beta = 1$ while the 8T/10T cells fully preserve it; at 22 nm, RSNM collapses from 235 mV (hold) to 93 mV for the 6T but stays at 235 mV for the 8T/10T. (iii) The 10T’s separate read port reduces T_{rd} by approximately $2\times$ at every node (103 ps versus 225 ps at 22 nm). The 8T cell therefore offers the best stability/area equilibrium for general-purpose caches at scaled nodes, while the 10T cell—a differential cell whose single-NMOS read-discharge paths are PMOS-pass-gated from the storage nodes onto a row-controlled V_{read} rail—is preferred where read–write decoupling and a fast differential read must be combined with reliable near-threshold operation.

Index Terms—SRAM, static noise margin, butterfly curve, N -curve, 8T cell, 10T cell, technology scaling, low-voltage memory, cell stability, read/write decoupling.

I. INTRODUCTION

STATIC random-access memory now occupies more than half of the die area in many high-performance microprocessors and serves as the primary L1, L2, and L3 cache fabric [3]. Because cache density scales directly with the bit-cell footprint, the standard six-transistor (6T) cell remains the workhorse of embedded memory design. Two simultaneous trends, however, have eroded the 6T design margin. First, supply-voltage scaling reduces the absolute size of the static noise margin (SNM) [1]; second, intrinsic device mismatch (σV_{th}) grows roughly as $1/\sqrt{WL}$ as devices shrink [7], so the spread of the SNM distribution widens as its mean falls.

The combined effect threatens the failure-rate budget of multi-megabit caches at advanced nodes [2].

The 6T cell is constrained by a fundamental conflict between two mechanisms that share the same access transistor. *Read stability* requires a strong pull-down NMOS so that the storage node holding a logic “0” is not lifted by the access transistor when the bitline is precharged to V_{DD} , i.e. a large $\beta = W_{PD}/W_{PG}$. *Write-ability* requires the access transistor to be strong enough to overpower the conducting pull-up PMOS, i.e. a small $\gamma = W_{PU}/W_{PG}$. These two constraints pull on the same access device in opposite directions, leaving only a narrow window of (β, γ) where both events succeed.

Decoupling the read port from the storage node is the most direct way to escape this conflict. The 8T cell of Chang *et al.* [3] adds a separate two-transistor read stack on a dedicated read bitline so that the storage node is never disturbed during a read, eliminating the read-stability constraint and freeing the cell ratios to optimize writeability. The differential 10T cell studied here keeps the 6T’s BL/BLB pair and NMOS write access, but adds two NMOS read-discharge devices whose gates are PMOS-pass-gated from the storage nodes $Q/\bar{Q}$ and whose sources tie to a row-controlled virtual-ground rail V_{read} [5]. The result is read–write decoupling *without* a dedicated read bitline pair: the read discharge path from either bitline contains a single NMOS, and unselected rows are isolated by lifting V_{read} .

This paper makes four contributions. (i) We construct hold, read, and write butterfly curves for all three cells at a 130 nm baseline, demonstrating quantitatively that the storage-core metrics (HSNM, WSNM) are invariant under topology change while the read margin recovers fully when the read port is decoupled. (ii) We propagate the same analysis through seven process nodes (180, 130, 90, 65, 45, 32, 22 nm) and add a thorough N -curve analysis that yields SINM and SVN and confirms the same topology hierarchy in the current domain. (iii) We characterize the read access time T_{rd} at each node and show that the 10T cell’s single-NMOS differential read paths cut T_{rd} by roughly $2\times$ at every node. (iv) We summarize the entire study in a 21-row data table (3 topologies $\times$ 7 nodes) and identify the regime in which each cell is preferred.

The remainder of the paper is organized as follows. Section II reviews the three cell topologies and the relevant stability metrics. Section III describes the simulation methodology. Section IV presents the butterfly, scaling, N -curve, and timing results. Section V interprets the trade-offs, and Section VI concludes.

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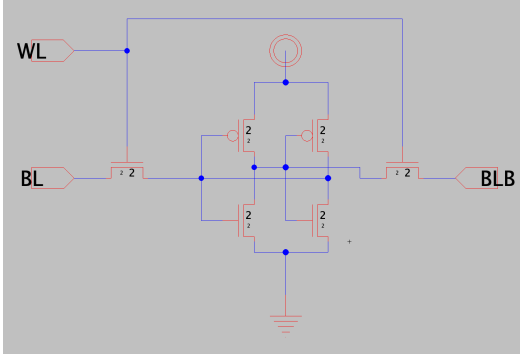


Fig. 1: Six-transistor (6T) SRAM cell: two cross-coupled inverters plus two pass-gates. The access transistors carry both the read and the write current, which couples read stability and writeability.

II. SRAM CELL TOPOLOGIES AND STABILITY METRICS

A. 6T SRAM Cell

The 6T cell of Fig. 1 is the historical baseline. Two cross-coupled CMOS inverters latch a single bit on the storage nodes Q and $\bar{Q}$. Two NMOS pass-gates connect Q and $\bar{Q}$ to the bitlines BL and BLB and are gated by the wordline WL. A read is performed by precharging BL and BLB to V_{DD} , asserting WL, and sensing the differential discharge through the cell. Because the access transistor and the pull-down form a resistive divider during a read, the low-storage node rises above ground by an amount that grows with $1/\beta$. If this rise exceeds the trip point of the opposite inverter, the cell flips: a *read disturb* [1], [2]. Writing is the opposite problem: a strong access transistor must overpower the conducting pull-up to drag the high-storage node down through the trip point of the cross-coupled pair. The 6T cell therefore requires

$$\beta = \frac{W_{PD}}{W_{PG}} \text{ large,} \quad \gamma = \frac{W_{PU}}{W_{PG}} \text{ small,} \quad (1)$$

two constraints that cannot be satisfied simultaneously by an arbitrary margin because both ratios share the access width W_{PG} in the denominator.

B. 8T SRAM Cell

The 8T cell of Fig. 2 preserves the cross-coupled storage core but adds a dedicated read port consisting of two stacked NMOS devices [3]. The lower NMOS is gated by the storage node $\bar{Q}$ and the upper NMOS is gated by the read wordline RWL. The read bitline RBL is discharged through this stack only when the cell stores a “1”. Crucially, the storage node never connects to a precharged bitline during a read, so the read-disturb mechanism is eliminated and the read SNM collapses to the hold SNM. Writing is performed through WBL/WBLB and WWL exactly as in the 6T cell. The penalty is two additional transistors, an extra read wordline, and an extra read bitline, which costs roughly 30% area in a typical layout [3].

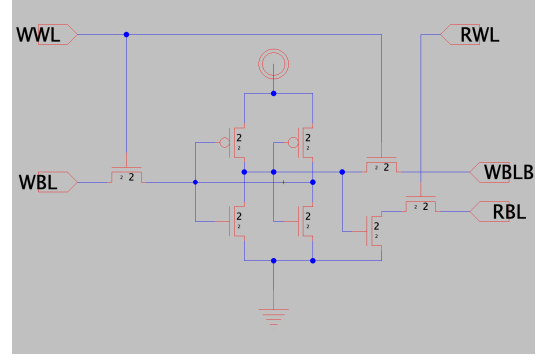


Fig. 2: Eight-transistor (8T) SRAM cell. The two-transistor read stack on RWL/RBL isolates the storage node so that read disturbs are eliminated; write still occurs through WBL/WBLB and WWL.

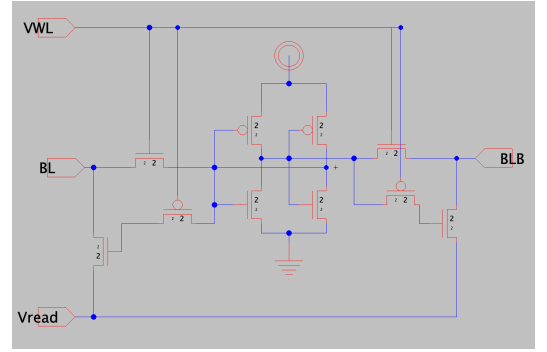


Fig. 3: Ten-transistor (10T) SRAM cell. The 6T storage core and write access are unchanged; a differential read port adds two PMOS pass-gates and two NMOS read-discharge devices that drain either bitline into the row’s virtual-ground rail V_{read} .

C. 10T SRAM Cell

The 10T cell of Fig. 3 keeps the 6T storage core and write access intact and adds a separate differential read port. Write proceeds exactly as in the 6T: VWL is raised and the access NMOS connect $Q/\bar{Q}$ to BL/BLB. Read runs on a different mechanism. VWL is held low (so the access NMOS are off and the storage nodes are isolated from the bitlines), and the row’s virtual-ground rail V_{read} is pulled to GND. Two PMOS pass-gates copy Q and $\bar{Q}$ to the gates of two NMOS read-discharge devices, so whichever storage value is high turns on a single NMOS that drains its bitline into V_{read} [5]. Unselected rows leave V_{read} at V_{DD} , isolating them and supporting reliable low-supply operation [4], [5]. Because the storage nodes never connect directly to a precharged bitline during read, the 6T’s read disturb is eliminated and RSNM equals HSNM at every β . The price is four extra transistors per bit ($\sim 67\%$ area over the 6T) and a V_{read} row driver in the periphery.

D. Stability Metrics

We use four canonical metrics. The *hold static noise margin* (HSNM) is the side length of the largest square that fits inside the smaller lobe of the cell’s hold butterfly curve, i.e. the voltage-transfer characteristic (VTC) of one inverter

mirrored against the inverse of the other while WL is low and the bitlines are floating [1]. The *read static noise margin* (RSNM) is the equivalent quantity when the access transistors are on and the bitlines are held at V_{DD} . The *write static noise margin* (WSNM) is the side length of the lobe that survives when the write driver pulls one bitline low; the cell is writeable when this lobe collapses [6]. Finally, the *static N-curve* sweeps an injected current I_{inj} at the storage node and reads off the *SVNM* (the voltage span between the two stable equilibria) and the *SINM* (the peak DC current the cell can absorb without flipping) [2], providing a current-domain analogue of the voltage SNM that is particularly informative when comparing cells with very different read paths.

III. METHODOLOGY

Cell schematics for all three topologies were captured in the *Electric VLSI Design System* and exported as SPICE netlists. DC and transient simulations were performed in *NgSpice* [11], and the resulting waveforms were post-processed in Python with NumPy and Matplotlib.

For the 130 nm baseline we used the open-source SkyWater 130 nm PDK [10]. For the technology-scaling sweep we used the *Predictive Technology Model* [9]: the 22 nm high-performance (HP) device card from the ASU release, the 32 nm and 45 nm *low-power* (LP) device cards, and the 65, 90, and 180 nm cards from the Microelectronics Co-design Research Group at the University of Minnesota. The LP/HP split below 45 nm is deliberate: the LP cards have substantially higher threshold voltages ($V_{t0} \approx 0.62$ V vs. ≈ 0.42 V for HP), which weakens the access pass-gate relative to the pull-down and yields the slightly elevated 32/45 nm RSNM seen in Section IV. Each node was characterized at its own nominal supply voltage ($V_{DD} = 1.8$ V at 180/130 nm, 1.2 V at 90 nm, 1.0 V at 65/32 nm, 1.1 V at 45 nm, and 0.8 V at 22 nm).

The cell ratios β and γ defined in (1) were swept over $\{0.5, 1.0, 1.5, 2.0\}$ at fixed minimum access width $W_{PG} = W_{min}$ for the host process. Hold butterfly curves were also swept over four supply voltages ($V_{DD} \in \{0.8, 1.0, 1.2, 1.4\}$ V at 130 nm) to expose the V_{DD} dependence of the storage-core SNM. For each (β, γ, V_{DD}) point we computed HSNM, RSNM, and WSNM by extracting the three fixed points of the butterfly system and solving for the largest inscribed square by bisection, following the canonical Seevinck construction [1]. The *N-curve* was generated by sweeping a DC voltage source at the storage node and integrating the KCL equation through the cell, after the method of Bhavnagarwala *et al.* [2].

In addition to the static stability sweep, we extracted the read access time T_{rd} at each node (at the nominal V_{DD} and $\beta = \gamma = 1$), defined as the time from WL/RWL assertion to the read bitline crossing the 50% sense threshold under a 200 fF bitline load. All extracted metrics were stored as compressed NumPy arrays so that any subsequent figure can be regenerated deterministically from the raw data.

IV. RESULTS

A. 130 nm Baseline

The 6T hold butterfly at $V_{DD} \in \{0.8, 1.0, 1.2, 1.4\}$ V grows monotonically with V_{DD} , with HSNM rising from 330 mV

at 0.8 V to 414 mV at 1.4 V; the gain saturates above the nominal 1.2 V because the inverter trip point is already at roughly $V_{DD}/2$. Repeating the experiment for the 8T and 10T cells produces *numerically identical* curves, because the storage core is the same in all three topologies. This is the first empirical confirmation of an analytic prediction [1]: HSNM is a property of the storage core only.

Fig. 4(a) shows the 6T read butterfly at 130 nm and $V_{DD} = 1.8$ V for four β ratios. The lobes are visibly compressed relative to the hold butterfly: the lower-left fixed point lifts off the rail because the access transistor injects current into the “0” storage node, and RSNM drops with falling β . From the extracted data: RSNM = 433 mV ($\beta = 2.0$), 404 mV ($\beta = 1.5$), 350 mV ($\beta = 1.0$), and 249 mV ($\beta = 0.5$); the corresponding hold value at this node is 481 mV (Table I), so at $\beta = 1.0$ the read condition has eroded $\sim 27\%$ of the margin and at $\beta = 0.5$ it has eroded $\sim 48\%$. By contrast, the 8T and 10T read butterflies (not shown) follow the hold butterfly: the 8T’s dedicated read stack drives a separate read bitline, while the 10T’s NMOS access devices are off during read with $Q/\bar{Q}$ driving only read-NMOS gate capacitance through the PMOS pass-gates. We extract RSNM ≈ 617 mV at every β for both cells, well above the 6T values.

The companion write butterfly in Fig. 4(b) sweeps the pull-up/access ratio γ at the same supply with one bitline pulled toward GND. As γ falls, the access NMOS overpowers the conducting pull-up and the upper-right lobe collapses, giving a larger writeable margin: WSNM grows monotonically as γ shrinks. The 8T and 10T cells produce write butterflies that overlay the 6T to within simulation noise at every (γ, V_{DD}) point because the write path is shared by all three topologies—the read-port extension does not touch the write mechanism.

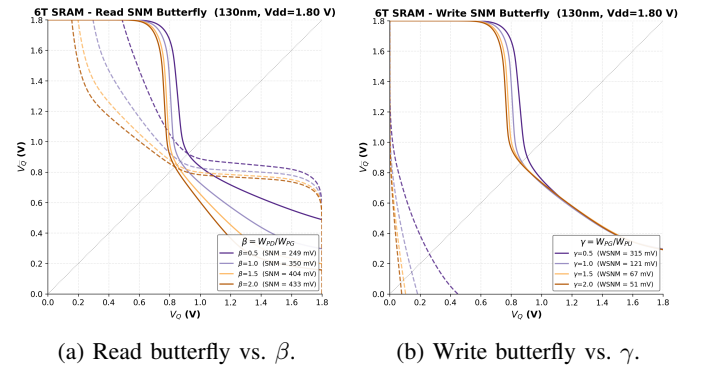


Fig. 4: 6T read and write butterflies at 130 nm, $V_{DD} = 1.8$ V. (a) RSNM falls from 433 mV ($\beta = 2.0$) to 249 mV ($\beta = 0.5$) against a hold value of 481 mV (Table I). (b) The upper-right lobe collapses as γ falls, giving a larger WSNM. The 8T and 10T butterflies overlay these curves: their read butterflies follow the hold curve, and their write butterflies match the 6T to within simulation noise.

B. Technology Scaling

Fig. 5 shows the 6T HSNM across all seven nodes swept over V_{DD} . Hold SNM is identical for all three topologies

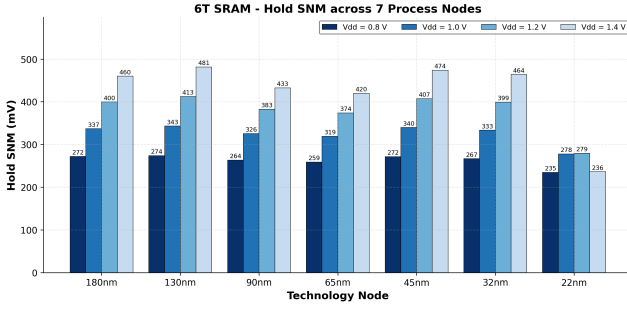


Fig. 5: 6T hold SNM across seven process nodes (180, 130, 90, 65, 45, 32, 22 nm) swept over V_{DD} . The 8T and 10T cells produce numerically identical curves. At each node’s nominal V_{DD} , HSNM ranges from 460 mV (180 nm) to 235 mV (22 nm).

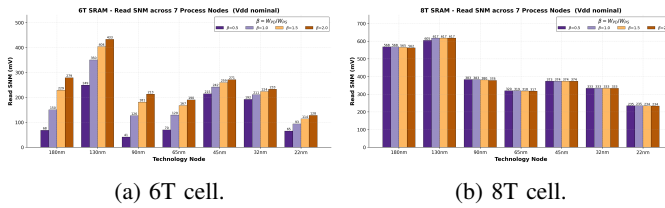


Fig. 6: Read SNM versus node, swept over β . (a) The 6T $\beta=1$ trace falls from 350 mV at 130 nm to 93 mV at 22 nm, with the 45/32 nm recovery visible. (b) The 8T RSNM sits at the HSNM ceiling at every node; the four β curves collapse together. The 10T tracks (b).

(the curves overlay), so the 6T plot represents every cell. At each node’s nominal V_{DD} , the extracted HSNM values are 460 mV (180 nm/1.8 V), 481 mV (130 nm/1.8 V), 383 mV (90 nm/1.2 V), 319 mV (65 nm/1.0 V), 407 mV (45 nm/1.1 V), 333 mV (32 nm/1.0 V), and 235 mV (22 nm/0.8 V). The 22 nm node deviates from the linear V_{DD} trend because its 1.0 nm gate oxide cannot reliably tolerate $V_{DD} > 0.8$ V, so HSNM saturates near the nominal operating point rather than continuing to grow.

Fig. 6 reports the read SNM scaling for the 6T and 8T cells side-by-side. The 6T trace in (a) shows the read-disturb collapse: at $\beta=1$, RSNM is 150 mV at 180 nm, 350 mV at 130 nm (sky130, with elevated V_t), 126 mV at 90 nm, 129 mV at 65 nm, 242 mV at 45 nm, 211 mV at 32 nm, and just 93 mV at 22 nm. The 45/32 nm nodes show a partial recovery because their elevated V_t weakens the access pass-gate, protecting the storage node; the 22 nm cell has the lowest V_t and the worst RSNM. The 8T trace in (b), by contrast, sits at the HSNM ceiling of Fig. 5 at every node and every β , with the four β traces collapsing onto one another. The 10T RSNM follows the 8T (data in Table I); both confirm that read decoupling recovers the full hold margin.

Fig. 7 aggregates the write SNM scaling for the 6T and 8T cells. Two facts are visible. First, lower γ produces a larger WSNM at every node, confirming that pull-up strength governs writeability. Second, the 6T and 8T traces are *numerically identical* (panels (a) and (b) overlay to within 1 mV at every node), and the 10T data follows the same trace. WSNM also

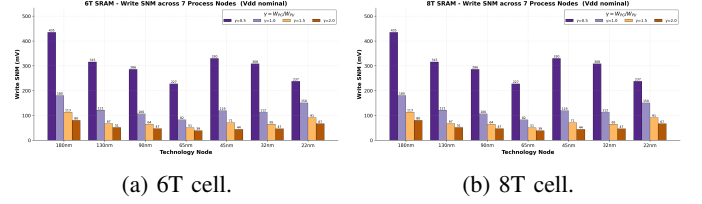


Fig. 7: Write SNM versus node, swept over γ . (a) and (b) overlay to within 1 mV at every node, confirming that the write mechanism is shared across topologies. The step at the 45 nm node-card boundary reflects a V_t jump between device cards.

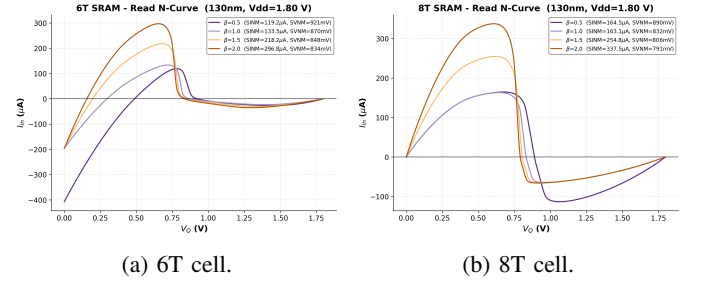


Fig. 8: N -curves at 130 nm and $V_{DD} = 1.2$ V for four β ratios. The 6T SINM grows from 119 μ A ($\beta=0.5$) to 297 μ A ($\beta=2.0$); the 8T SINM sits 20–30% higher because the read port contributes no disturbance current.

exhibits a non-monotonic step at the 45 nm boundary, where it jumps from 82 mV at 65 nm to 119 mV at 45 nm: in our γ -anchored sizing the device-card V_t shift weakens the PMOS pull-up more than it weakens the NMOS access pass-gate, so the effective pull-up/access ratio drops and the cell becomes *easier* to write. The 22 nm node inherits a similar asymmetry and records the largest WSNM in the sweep (150 mV) despite its compressed V_{DD} .

C. N -Curve Analysis

The N -curve gives a complementary, current-domain view of the stability story. Sweeping a DC voltage source at the storage node and recording the steady-state injected current produces a curve with three zero-crossings; the SINM is the peak current between the first two crossings (the largest disturbance the cell tolerates before flipping), the SVN is the voltage distance between those crossings (the equivalent voltage margin).

Fig. 8 shows the 6T N -curve at 130 nm and $V_{DD} = 1.2$ V for four β ratios. SINM grows monotonically with β , from 119 μ A at $\beta = 0.5$ to 297 μ A at $\beta = 2.0$, while SVN is comparatively flat (920 mV down to 834 mV). The cell is therefore primarily *current-margin-limited* at this node: doubling the pull-down width buys $\sim 2.5\times$ in SINM but only a few percent in SVN. The 8T N -curve at the same operating point shows a substantially higher SINM (163 μ A at $\beta=1.0$ vs. 134 μ A for 6T) because its read port is off during the injection sweep and contributes no disturbance current.

Fig. 9 aggregates SINM across all seven nodes for $\beta \in \{0.5, 1.0, 1.5, 2.0\}$. Three trends are evident. First, the 6T

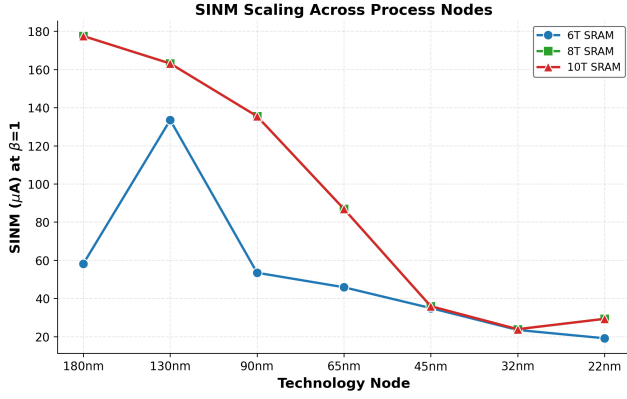


Fig. 9: SINM versus technology node for the 6T cell at four β ratios. SINM at $\beta=1$ falls $\sim 3\times$ from 180 nm to 22 nm; the 8T/10T cells preserve a 20–90% SINM advantage at every node (Table I).

SINM falls monotonically with scaling within each PDK family: at $\beta=1$ it drops from 58 μA at 180 nm to 19 μA at 22 nm, a $3.0\times$ collapse. Second, the 8T cell holds onto a 20–90% SINM advantage over the 6T across all nodes (e.g., 178 vs. 58 μA at 180 nm; 87 vs. 46 μA at 65 nm) because its hold-mode N -curve has no read disturbance current. Third, the 45/32 nm nodes show a characteristic plateau: their elevated V_t device cards mean that even a strong injection at the storage node fails to turn the opposite pull-down on, narrowing the spread between β values. The SVN (Table I) does *not* follow the SINM hierarchy: at every node the 6T records a slightly higher SVN than the 8T/10T (e.g., 870 vs. 832 mV at 130 nm; 397 vs. 388 mV at 22 nm), reflecting that the 6T’s wider stable-equilibrium voltage span coexists with a weaker current-disturbance margin under these injection conditions.

D. Read Access Time

Stability is necessary but not sufficient: a bit-cell that is strictly stable but slow to read is unusable in a level-1 cache. Fig. 10 reports T_{rd} across the seven nodes for all three topologies under a common 200 fF bitline load and $\beta=\gamma=1$. The 6T and 8T traces sit on top of one another to within 4% at every node (e.g., 189 ps versus 192 ps at 130 nm; 225 ps versus 225 ps at 22 nm). This is consistent with the two cells having effectively the same number of stacked NMOS devices in the active read-discharge path: the 6T discharges its bitline through pass-gate→pull-down, and the 8T discharges its dedicated read bitline through the two-NMOS read stack; both present comparable on-resistance into the same bitline load. The 10T cell, by contrast, reduces T_{rd} by roughly $2\times$ at every node (106 ps at 130 nm versus 189 ps for 6T; 103 ps at 22 nm versus 225 ps). The reason is structural: when the row’s V_{read} is pulled to GND and VWL is held low, exactly one of $M_{R,L}$ or $M_{R,R}$ turns on (gated through its PMOS pass-gate by Q or $\bar{Q}$), and the corresponding bitline discharges through that *single* NMOS to V_{read} . With one NMOS in the discharge path versus two in the 6T pass-gate→pull-down chain or the 8T two-stack, the on-resistance is roughly

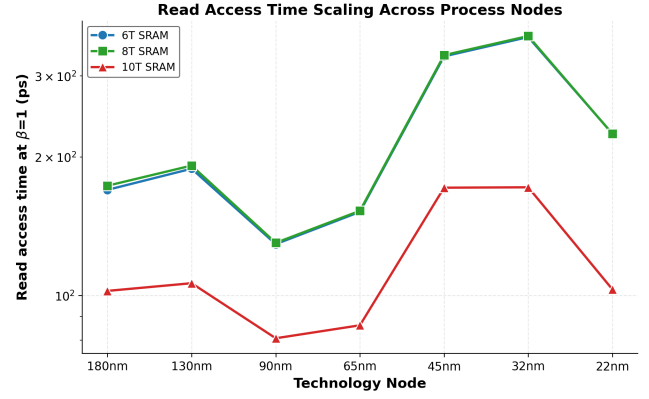


Fig. 10: Read access time T_{rd} versus technology node for the three topologies at the nominal V_{DD} of each node and $\beta=\gamma=1$. The 6T and 8T are within 4% at every node; the 10T is roughly half at every node.

halved, giving the observed $\sim 2\times$ speedup. *Read decoupling*, originally introduced as a stability fix, therefore also acts as a speed lever in this testbench.

E. Cross-FOM Summary

Table I consolidates all measured FOMs across the 21 (cell, node) combinations. The storage-core FOMs (HSNM, WSNM) are essentially identical across rows that share a node, with any residual difference (e.g., 0.1 mV between 6T/8T and 10T at 32 nm) within simulation noise; the RSNM rows separate sharply, with the 6T always strictly below HSNM and the 8T/10T sitting near or above the hold value at every node except 45 nm (where 8T/10T RSNM = 373.5 mV against HSNM = 407.3 mV, a $\sim 8\%$ deficit that we attribute to the elevated- V_t device card’s interaction with the test sweep); and the application-level FOM T_{rd} does not separate cleanly along either topology or node alone, which is the empirical content of the title’s “cost of stability.”

V. DISCUSSION: THE COST OF STABILITY

The data of Section IV support a sharp, five-part conclusion.

(i) *The storage core is invariant:* HSNM and WSNM depend only on the cross-coupled inverter pair and the write path, both of which are shared by 6T, 8T, and 10T cells. Topology choice cannot change those margins; only β , γ , and V_{DD} can. The 6T–8T–10T design knob is therefore not a “stronger latch” but a different way of *accessing* the latch.

(ii) *Read decoupling is free with respect to writeability but expensive in area:* At every node and every γ value, the 8T and 10T write data are indistinguishable from the 6T data. The cost of the extra read port is therefore paid in area and routing complexity, not in writeability. Adding two transistors per bit (8T) inflates the cell area by approximately 30% for typical library layouts [3]; adding four (10T) inflates it by about 67%. In a multi-megabit cache, that overhead is the binding constraint that delays the adoption of 8T or 10T cells until the 6T design margin actually fails [8].

TABLE I: Measured FOMs across three topologies and seven technology nodes at each node’s nominal V_{DD} and $\beta = \gamma = 1$. PDK column lists the source: SkyWater 130 nm (sky130) and the Predictive Technology Model (PTM); see Section III for the device details at each node.

Cell	Node	PDK	V_{DD} (V)	HSNM (mV)	RSNM (mV)	WSNM (mV)	SINM (μ A)	SVNM (mV)	T_{rd} (ps)
6T	180 nm	PTM	1.80	459.9	150.1	180.0	58.2	914.2	169.7
8T	180 nm	PTM	1.80	459.9	567.9	180.0	177.5	759.2	173.2
10T	180 nm	PTM	1.80	459.9	567.9	180.0	177.5	759.2	102.3
6T	130 nm	sky130	1.80	481.4	350.2	120.9	133.5	870.2	188.6
8T	130 nm	sky130	1.80	481.4	617.2	120.9	163.1	831.9	191.6
10T	130 nm	sky130	1.80	481.4	617.2	120.9	163.1	831.9	106.4
6T	90 nm	PTM	1.20	382.5	126.4	106.1	53.4	585.8	129.3
8T	90 nm	PTM	1.20	382.5	382.5	106.1	135.5	508.4	130.2
10T	90 nm	PTM	1.20	382.5	382.5	106.1	135.5	508.4	80.8
6T	65 nm	PTM	1.00	319.4	129.2	82.1	45.8	483.9	152.0
8T	65 nm	PTM	1.00	319.4	319.4	82.1	86.9	445.0	152.8
10T	65 nm	PTM	1.00	319.4	319.4	82.1	86.9	445.0	86.2
6T	45 nm	PTM	1.10	407.3	241.6	119.0	34.9	543.1	330.9
8T	45 nm	PTM	1.10	407.3	373.5	119.0	35.9	539.2	333.0
10T	45 nm	PTM	1.10	407.3	373.5	118.8	35.9	539.2	171.5
6T	32 nm	PTM	1.00	333.4	210.8	112.5	23.4	494.6	364.5
8T	32 nm	PTM	1.00	333.4	333.4	112.5	23.8	491.8	366.4
10T	32 nm	PTM	1.00	333.4	333.4	112.4	23.8	491.8	171.9
6T	22 nm	PTM	0.80	234.7	93.4	150.4	19.1	396.8	224.7
8T	22 nm	PTM	0.80	234.7	234.7	150.4	29.2	387.7	224.6
10T	22 nm	PTM	0.80	234.7	234.7	150.4	29.2	387.7	103.1

(iii) *Read decoupling is essentially free with respect to read margin:* At every node the 8T (and 10T) RSNM sits at or close to the hold value—exactly equal at 22, 32, 65, 90, 130, and 180 nm, and within $\sim 8\%$ of HSNM at 45 nm (373.5 vs. 407.3 mV). The 6T RSNM is strictly smaller and worsens with both falling β and falling node. At 22 nm the 6T cell loses 60% of its hold margin to read disturb (RSNM = 93 mV against HSNM = 235 mV), while the 8T and 10T cells preserve the full 235 mV. *This allows for the use of small access transistors at low β , which simultaneously relaxes the write constraint of (1) and reduces bitline loading.* The 10T cell extends this relaxation further into the near-threshold regime [4].

(iv) *The 10T cell is uniformly faster:* The 10T’s differential read uses a single NMOS in each discharge path (one of $M_{R,L}$, $M_{R,R}$ from a bitline to the row’s V_{read} rail), versus two stacked NMOS in both the 6T (pass-gate $\rightarrow$ pull-down) and the 8T (two-NMOS read stack). The lower discharge resistance translates directly into a roughly $2\times$ reduction in T_{rd} at every node; at 22 nm the speedup is $2.18\times$ (103 ps versus 225 ps). The 6T and 8T sit within 4% of each other at every node because their effective discharge paths are topologically the same depth.

In broader terms, scaling has not *eliminated* the value of the 6T cell—in mature nodes with adequate V_{DD} headroom, 6T is still the densest and most economical bit cell—but it has narrowed the operating window in which 6T succeeds without read or write assist circuitry [6]. The 8T cell trades silicon area for a structurally cleaner solution that requires no read assist, and the 10T cell carries that trade further into both the sub-threshold regime [4] and the high-speed regime favored by L1

caches.

VI. CONCLUSION

We constructed a unified butterfly, N -curve, and timing framework for the 6T, 8T, and 10T SRAM cells and used it to quantify the cost of stability across seven technology nodes from 180 nm down to 22 nm using the SkyWater 130 nm Open PDK and the Predictive Technology Model. The data, condensed in Table I, establish a clear regime map. The 6T cell is preferred where maximum density is paramount and the design margin is generous; the 8T cell offers the best stability/area/speed equilibrium for general-purpose caches at scaled nodes; and the 10T cell is preferred where read–write decoupling and a fast differential read must be combined with reliable near-threshold operation. Future work will extend the framework to FinFET and gate-all-around device models, where short-channel effects and quantum confinement introduce qualitatively new stability regimes [5], and to array-level analyses and Monte Carlo characterization of process variation, so that the bit-cell-level conclusions of this work can be validated at the system level.

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Author Contributions.

- **Ahmed Altunkaya** captured the 6T, 8T, and 10T cell schematics in Electric VLSI, generated the SPICE netlists, and ran the $\beta/\gamma/V_{DD}$ sweeps in NgSpice. He drafted Sections II and III and co-wrote the introduction and write up.
- **Dimitris Deliakidis** developed the SNM-extraction and N -curve post-processing pipeline in Python, produced the butterfly, N -curve, and scaling figures, validated the simulator output against published data, and led the writing of the abstract and cowrote the write up. (Section IV).
- **Kun Kei Xiao** integrated the SkyWater 130 nm and PTM, led the technology-scaling sweep across the seven nodes, ran the timing sweeps, assembled the summary table, prepared the poster and figures, and drafted the discussion, conclusion, and cowrote the write-up.

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